

# Utku Egemen Umut

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## EDUCATION

**University of Toronto - Computer Science (cGPA: 3.99 / 4.00)**

Sep 2022 – Jun 2027 (PEY)

*Honours Bachelor of Science, Computer Science Specialist and Math Minor*

*U of T Scholar; Trinity Scholarships*

**Coursework:** ML, Deep Learning, Parallel & Systems Programming, Computer Architecture, Operating

Systems, Computer Networks, Distributed Systems, Probabilistic Learning, Computer Vision, ML Engineering

## EXPERIENCE

**Applied Machine Learning Engineer Intern (PEY, 16 months)**

May 2025 – Aug 2026

*Cerebras Systems* 🔗

*Toronto, Canada*

- Extended **CePO** to vision-language inputs: threaded images through its multi-stage planner, fixed the alternating-message contract its prompts depend on, and added LLM rating and verification functions.
- Designed the **4-way distillation comparison** (GRPO, teacher-OPD, OPSD, SDPO) over difficulty-bucketed math and codegen data, and ran the failure-mode study showing **sparse-reward RL** both stalls on problems the model never solves and reinforces correct answers reached by incorrect reasoning; **authored the whitepaper**.
- Implemented **self-OPD** (OPSD and SDPO) in **verl**, **unifying four training methods** via one teacher-context switch and shared training/eval path. **Improved AIME24 Avg@12 from 48.6 to 53.3** by replacing a gradient-saturating, summed-per-token KL clamp with a **per-element forward-KL clip**; contributed to the **length-control white paper**.
- Owned **agentic infrastructure**: integrated **SkyRL** and shipped **GEPA** prompt optimization into Applied ML workflows (reported a 10% improvement over GRPO using 35× fewer rollouts).
- Ported the open-source **DSA** sparse-attention stack onto a **single 24 GB A10** by building an eager-mode load path with injected DSA config and checkpoint warmstart, fitting an **H100-class** memory profile onto one GPU.

**Software Engineer**

Jan 2025 – Apr 2025

*Knowlecye Technologies Inc.*

*Toronto, Canada*

- Developed a **Python/FastAPI backend** with **SQL-backed data access**, implementing REST APIs for posts, comments, content discovery, and topic-based tagging.
- Engineered an **event-driven notification microservice** using **gRPC** and **RabbitMQ**, enabling **asynchronous** processing of forum interactions.

**Computer Vision Researcher – LURA Research Award**

May 2024 – Aug 2024

*Computational Vision and Imaging Lab at York University, supervised by Kosta Derpanis* 🔗

*Toronto, Canada*

- Built a **scene-detection pipeline** (template matching and masking) achieving **91% accuracy** in selecting relevant frames from **1,000+** frames of televised footage, cutting false positives on irrelevant scenes.
- Implemented 2D **point tracking** robust to camera motion, supplying the environmental context that improved **3D human pose estimation**.

**Undergraduate Researcher**

Jun 2023 – Sep 2023

*Autonomous Vision Group at KUIS AI Research Centre, supervised by Prof. Fatma Guney* 🔗

*Istanbul, Turkey*

- Cut **multi-agent trajectory-prediction** runtime **30%** by rewriting the experiment pipeline for a **Linux HPC cluster**, raising iteration speed on the group's autonomous-driving models.

**Machine Learning Engineer Intern**

May 2023 – Aug 2023

*TAZI AI Systems* 🔗

*Istanbul, Turkey*

- Shipped **end-to-end churn-prediction models** into the **Adaptive AutoML** platform, achieving **89%** accuracy.

## OPEN SOURCE, TEACHING & LEADERSHIP

*Students Developing Software (SDS), University of Toronto, supervised by Prof. David Liu* 🔗

2023 – 2024

- MarkUs** 🌐 | *Ruby on Rails, React, PostgreSQL, Docker*: grading platform for **5,000+** students.
- PyTA** 🌐 | *Python, Pylint, Astroid*: static analysis for **5,000+** students annually.
- MemoryViz** 🌐 | *JavaScript, Node.js*: rebuilt a prototype into a maintainable codebase with thousands of users.
- Shipped every change through **code review** and **CI** with test suites, landing upstream in course tooling.

**Lead Tutorial TA**, CSC236 (2 semesters), UofT 🔗 | **ML Projects Director**, UTMIST 🔗

Sep 2024 – Present

## TECHNICAL SKILLS

**Languages:** Python, C++, C, SQL, Shell, Java, JavaScript, Haskell

**Backend & Databases:** FastAPI, gRPC, RabbitMQ, PostgreSQL

**ML:** PyTorch, FlexAttention, tensor parallelism, distributed training, vLLM, verl, HF Transformers, Inspect, scikit-learn

**Systems & Tools:** Linux, Docker, Slurm/HPC, GPU programming, Git, CI/CD